

MULTI-STAGE AI PROMPT WORKFLOW

Smart Campus Maintenance Request Prioritization and Escalation

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1. Scenario Selected

This workflow helps a college maintenance team process unstructured complaints about campus facilities. It extracts key details, prioritizes requests based on safety, impact, urgency, repeated complaints, and waiting time, drafts an action plan, and then reviews and improves that plan using a mandatory critique stage.

Note: This is a training exercise in chained prompt design. It supports request organization and does not replace on-site inspection or the judgment of qualified maintenance personnel.

2. Objective

- Convert free-text maintenance complaints into structured requests.
- Assign a priority level based on safety risk, impact, urgency, repeated reports, and waiting time.
- Recommend a practical response and responsible team.
- Escalate unresolved high-priority requests that exceed their response limit.
- Demonstrate a three-step AI prompt chain with a mandatory self-critique stage.

3. Input Data Used

Ticket ID	Complaint / Presentation	Location	Repeat?	Waiting
M001	Water leaking from a ceiling near electrical equipment; floor is wet.	Library, 1st floor	No	10 min
M002	One classroom projector is not turning on before a scheduled lecture.	Block B, Room 204	No	35 min
M003	A stairwell light is out; students reported the same issue twice this week.	Block A, Stairwell 2	Yes—3rd report	25 min
M004	A broken desk causes mild inconvenience; several desks remain usable.	Block C, Room 101	No	50 min
M005	A restroom tap is broken and water is spreading across the floor.	Admin block, Ground floor	Yes—2nd report	40 min

4. Prompt Chain Transcript and Intermediate AI Responses

Step 1 — Extraction and Planning

Prompt 1 — Exact prompt used:

Act as a campus maintenance request analyst. Analyze the complaints below and extract the factors that determine priority, including safety risk, number of people affected, urgency, repeated reports, and waiting time. Create a structured prioritization framework and apply it to each ticket. Present a table with Ticket ID, Priority Level, Reason, Suggested Action, and Escalation Condition. Do not assume facts that are not provided. Input tickets: M001 ceiling water leak near electrical equipment; M002 classroom projector failure before a lecture; M003 stairwell light out, third report this week; M004 broken desk while other desks remain usable; M005 broken restroom tap with water spreading, second report.

Intermediate AI Response — Step 1:

Ticket	Priority	Reason	Suggested Action	Escalation Condition
M001	Critical	Water near electrical equipment may create an immediate safety hazard.	Keep people away; notify facilities/electrical staff urgently; inspect safely.	Escalate immediately if the area cannot be made safe.
M003	High	An unlit stairwell can create a fall risk; repeated reports suggest the issue remains unresolved.	Arrange prompt inspection and temporary safe lighting or access control.	Escalate if not addressed within 30 minutes or if risk increases.

Ticket	Priority	Reason	Suggested Action	Escalation Condition
M005	High	Water spreading on a restroom floor creates a slip risk, and this is a repeat report.	Restrict the wet area, place warning signs, and send plumbing staff.	Escalate if the leak is not contained within 30 minutes.
M002	Medium	The equipment issue affects a scheduled lecture but no direct safety risk is stated.	Check power/cables and arrange AV support or an alternative teaching setup.	Escalate if unresolved before the lecture starts.
M004	Low	A broken desk is inconvenient, but other desks are available and no immediate hazard is stated.	Log the repair and schedule maintenance.	Escalate if the desk creates a safety hazard or remains unresolved beyond the service target.

Step 2 — Generation / Drafting

Prompt 2 — Exact prompt used:

Using only the framework and ticket analysis from Step 1, create an operational response plan for the campus maintenance team. Do not introduce new priority categories. Explain how tickets are handled in priority order, how repeated reports affect urgency, how the team tracks unresolved requests, and when escalation occurs. Include a short notification message for the facilities manager when an escalation condition is reached.

Intermediate AI Response — Step 2:

The team should handle tickets in this order: Critical, High, Medium, then Low. Safety hazards are addressed first. Repeated reports increase attention because they may indicate that an issue remains unresolved. Each ticket should record its ID, location, report time, priority, repeat-report count, assigned team, status, and elapsed time. The queue should be checked regularly. If a ticket reaches its escalation condition without action, notify the facilities manager and update the ticket status.

Example escalation notification:

ALERT — Ticket M001. Location: Library, 1st floor. Issue: Water leaking near electrical equipment. Priority: CRITICAL. Waiting: 10 minutes. Status: Not yet made safe. Action required: Please dispatch the appropriate facilities/electrical team and confirm the area is safe.

Step 3 — Evaluation and Refinement (Critique Gate)

Prompt 3 — Exact prompt used:

Act as a strict senior reviewer. Evaluate the Step 2 plan using this checklist: (1) Are immediate safety hazards prioritized? (2) Are repeated reports explicitly used as an urgency signal? (3) Is tracking of unresolved tickets explained? (4) Does the escalation message include enough information to act? (5) Is the workflow practical for a campus maintenance team? First list weaknesses, then provide a corrected final version. Keep the original facts and priority framework; do not invent additional facts.

Intermediate AI Response — Step 3 Critique:

- The priority order is clear, but the plan should explicitly state that staff must keep people away from unsafe areas.
- Repeat reports are mentioned; the final version should state that they trigger review of the ticket history.
- The queue-checking process should include checking both ticket status and elapsed time.
- Notifications should include the location, hazard, priority, wait time, and required action.
- Staff should confirm when the hazard is contained or the ticket is resolved.

Refined Final Output:

The maintenance queue classifies tickets as Critical, High, Medium, or Low. Critical safety hazards are handled first, and staff should restrict access to unsafe areas while the appropriate team responds. High-priority tickets include unresolved hazards and repeated reports that may indicate a continuing problem. Repeated reports should be linked to the existing ticket history and reviewed for urgency. Medium and Low tickets are scheduled according to impact and available resources. The queue records the ticket ID, location, priority, repeat count, assigned team, status, and elapsed time. Staff regularly review open tickets and escalate them when their defined response limit is reached. The ticket is closed only after the responsible team confirms the issue has been addressed.

Final Notification Example:

ALERT — Unresolved Maintenance Ticket. Ticket ID: M003. Location: Block A, Stairwell 2. Issue: Stairwell light is out. Priority: HIGH. Repeat reports: 3rd report this week. Waiting: 25 minutes. Status: Unresolved. Action required: Please arrange prompt inspection and confirm the stairwell is safe.

5. Final Workflow Summary

Stage	Input	AI / System Action	Output
1. Extraction / Planning	Raw maintenance complaints	Extract risk, impact, repeats, and wait time; assign priority.	Prioritized ticket table
2. Generation / Drafting	Step 1 output	Create response plan, tracking rules, and escalation notice.	Operational plan + alert
3. Evaluation / Refinement	Step 2 output	Review against checklist and correct gaps.	Refined final workflow

6. Expected System Output

- A prioritized list of maintenance tickets.
- A Critical, High, Medium, or Low priority label.
- Detection and tracking of repeated reports.
- Assignment to a responsible maintenance team.
- Monitoring of open tickets and elapsed time.
- Escalation notifications when response limits are exceeded.
- A dashboard showing urgent and overdue tickets first.

7. Compliance with Assessment Rules

Requirement	Status
At least 3 chained prompts	Satisfied — three sequential prompts are documented.
Each prompt depends on previous output	Satisfied — Step 2 uses Step 1; Step 3 critiques and refines Step 2.
Critique gate	Satisfied — Step 3 evaluates the draft against a checklist.
Exact prompts and intermediate responses	Included in Sections 4 and 5.
Live chat evidence	Insert a screenshot from the actual AI interface if required by the assessment.

8. Conclusion

This multi-stage AI prompt workflow demonstrates how campus maintenance complaints can be organized through chained prompting rather than a single request. Step 1 extracts details and assigns priorities, Step 2 converts the analysis into an operational response plan, and Step 3 reviews and improves the plan through a critique gate. The workflow also tracks repeated reports and delayed action so that unresolved safety issues can be escalated to the facilities manager.